

Advanced International Macroeconomics B (EI136)

Class 5

Macroprudential policy, capital controls, integrated framework

Cédric Tille

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Outline of today class

- Macroprudential policy.
 - Pattern and effectiveness (Alam et al 2019).
 - Cushioning of adverse shocks (Bergeant 2020).
 - Spillovers beyond banks (Claessens et al. 2023).
- Use and effectiveness of capital controls.
 - Pattern from high frequency data (Binici, Das, and Pugacheva 2023).
 - Impact on capital flows (Chen 2024).
 - Surveys of the recent literature (Ma and Rebucci 2019; Erten, Korinek and Ocampo 2021).
- Integrated model with many frictions and tools.
 - Joint use and effect of complementary policies in a 3 period model (Cavallino and Hofmann 2022).
 - Extra slides at the end: analysis through a four-panel diagram in additional slides (Basu and Gopinath 2024).

MACROPRUDENTIAL POLICY

What is macroprudential policy?

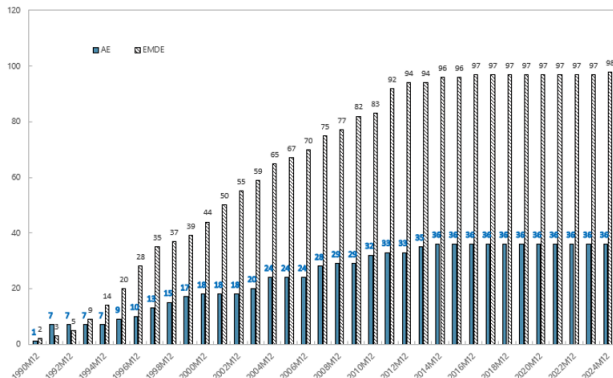
- Measures aimed at limiting leverage of borrowers or financial intermediaries (Alam et al. 2019).
 - Complement standard macroeconomic tools (interest rate) when dealing with volatile capital flows and financial intermediaries with frictions.
- Measuring the use of macroprudential policy is challenging (not a simple interest rate).
 - Dummy variables: if a measure is put in place (for instance higher capital requirements for banks) it is coded as +1 for all the years where it remains in place.
 - Limitation: does not reflect how strong the measure is (e.g. by how much the capital requirements are raised).
 - Assessing the strength of the measure would not be straightforward.
- Broad range of tools, some aimed at borrowers (debt-to-income ratio, loan-to-value limits), other aimed at intermediaries (leverage limits).

- IMF database with a broad coverage (Alam et al. 2019).
 - January 1990 to December 2016 (and ongoing updating), 134 countries (36 advanced, 98 emerging and developing). <https://www.elibrary-areaer.imf.org/Macroprudential/Pages/Home.aspx>
 - Dummy measures of whether a policy tool (17 in total) is tightened (+1) or relaxed (-1).
 - Includes measures of intensity for LTV measures at monthly frequency.
- Macroprudential policy usage has increased, and reached a plateau since 2012, with nearly all countries using them.
 - Advanced economies use mostly LTV limits, Reserves Requirements, and Capital adequacy.
 - Emerging use limits on Foreign Currency borrowing, RR, and liquidity requirements.

Growing number of countries using MaP

- Even among advanced economies.

Figure 1: Number of economies that have used macroprudential policy
(Except for reserve requirements)



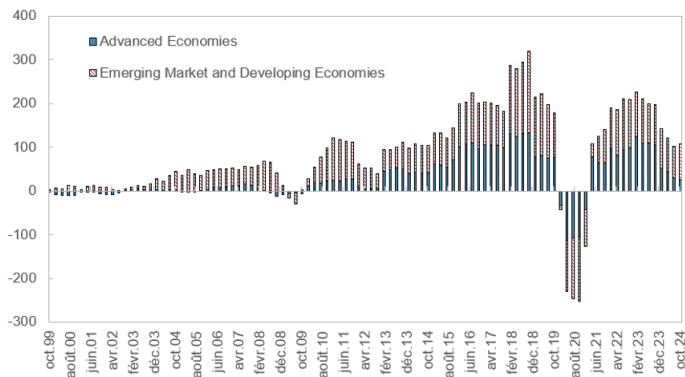
Alam, Zohair, Adrian Alter, Jesse Eisenman, Gaston Gelos, Heedon Kang, Machiko Narita, Elend Nier, and Naixi Wang (2019). "Digging Deeper – Evidence on the Effects of Macroprudential Policies from a New Database", IMF Working Papers 19/66. Update using: <https://www.elibrary-areaer.imf.org/Macroprudential/Pages/Home.aspx>

Tightening and easing

- Tightening from 2012 to 2020, and large easing during the Covid episode.

Figure 4: Usage of macroprudential policies over time

The bars indicate the cumulative sum of the net number of tightening actions of any macroprudential policy instruments over the current and past three quarters.



Alam, Zohair, Adrian Alter, Jesse Eisenman, Gaston Gelos, Heedon Kang, Machiko Narita, Elend Nier, and Naixi Wang (2019). "Digging Deeper – Evidence on the Effects of Macroprudential Policies from a New Database", IMF Working Papers 19/66. Update using: <https://www.elibrary-arear.imf.org/Macroprudential/Pages/Home.aspx>

Does it work?

- Regression of credit growth on macroprudential measures and controls (63 countries, incl. 29 emerging, with quarterly data).
- Macroprudential measures reduce household credit, especially in emerging countries.
 - Loan targeted measures (on borrowers or lenders) reduce credit in all countries.
 - LTV and DTI (debt service have strong effects).
 - Measures targeted at lenders have a stronger effect (loan provision, restrictions).
- Effect on house prices limited to advanced economies, mostly from borrower measures (LTV).
- Policies have side effects. Loan targeted measures reduce private consumption in advanced economies, but not GDP.

- Overall macroprudential policy lowers credit.
- Loan targeted measures are most effective (limits on debt service, loan to value).

Table 1. Summary: The Effects of Macroprudential Policies on Household Credit and House Prices

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable:	Real Household Credit (yoy growth)			Real House Prices (yoy growth)		
Group:	ALL	AE	EMDE	ALL	AE	EMED
MaPP Index (all tools)	-0.842***	-0.257	-1.388***	-0.0877	-0.898**	0.0619
Loan-Targeted	-1.883***	-1.043**	-2.925***	-1.066**	-1.667**	-0.615
Demand	-1.994***	-0.607	-4.926***	-1.406**	-2.049**	-0.213
Supply-All	-0.824*	0.395	-1.511***	0.253	0.0116	0.00789
Supply- Loans	-2.931***	-3.028**	-3.005**	-1.191	-1.970	-1.191
Supply- General	-0.602	0.958	-1.354*	0.485	0.0313	0.175
Supply- Capital	-1.009	0.221	-1.959*	0.270	0.0689	0.459
dsti	-2.648***	-1.231	-6.173**	-1.455	-3.167	2.537
ltv	-2.557***	-0.654	-5.797***	-1.952***	-2.627***	-1.020
tax	-0.988	-1.514	-0.860	-0.597	-5.159**	3.235***
lcr	-4.620	-2.142	-6.562*	-1.965	-0.0586	-2.436
lip	-5.865***	-4.161	-6.656***	-2.987**	-1.510	-4.529**
loanr	-3.489**	-4.592**	-2.885	-1.385	-4.164*	-0.262
capital	-1.091	0.0547	-1.803	0.304	-0.157	0.582
Observations	4,492	2,858	1,634	4,111	2,780	1,331
N (countries)	63	34	29	55	34	21
R-squared (avg)	0.884	0.873	0.903	0.812	0.846	0.786

Sources: The iMaPP database, Bloomberg, BIS, OECD, others (see Appendix IV), and authors' estimation.

Notes: The table reports the cumulative effects of the specified macroprudential tightening after four quarters, obtained by the fixed effects (FE) estimation. Each MaPP index/policy is considered individually. "dsti"= limits to the debt-service-to-income ratio, "ltv"= limits to the loan-to-value ratio, "tax"= tax measures, "lcr"= limits to credit growth, "lip"= loan loss provisions, "loanr"= loan restrictions, and "capital"= capital requirements. "MaPP index" is the sum of dummies for all of 17 categories. "Loan-targeted" group consists of the "Demand" and the "Supply-loans" instruments. "Demand": LTV and DSTI. "Supply-loans": lcr, lip, loanr, limits to the loan to deposit ratio, and limits to foreign currency loans. "Supply-general": reserve requirements, liquidity requirements, and limits to FX positions. "Supply-capital": leverage, countercyclical buffers, conservation buffers, and capital requirements. See Appendix I Table 3 for the definitions of instruments. For each country group, R-squared is computed as the average of the individual regressions. AE = advanced economies; and EMDE = emerging market and developing economies.

Alam, Zohair, Adrian Alter, Jesse Eisenman, Gaston Gelos, Heedon Kang, Machiko Narity, Elend Nier, and Naixi Wang (2019). "Digging Deeper – Evidence on the Effects of Macroprudential Policies from a New Database", IMF Working Papers 19/66.

CUSHIONING EFFECT

Macroprudential policy as a cushion

- Policy may not have a direct effect, but an indirect one by limiting the impact of adverse shocks.
- Does a tighter level of macroprudential policy limits the impact of shocks on GDP (Bergant et al. 2020)?
 - Shocks: increase in Vix, higher U.S. interest rate, higher net capital outflows (sudden stop, IV with overall net outflows for emerging economies).
 - Impact of macroprudential in level and interaction with shock.
- More prudential policy limits the impact of shocks on growth.
 - For Vix and capital flows shocks.
 - Decreasing returns to scale in the effect.
- More prudential policy provides monetary policy autonomy (less increase in interest rate following a higher Vix or sudden stop).
- No evidence of negative international spillovers.

- Adverse shocks reduce growth.
- Tighter macroprudential limits the impact, with decreasing returns.

Table 1: Dampening Effects of Macroprudential Regulation on GDP Growth

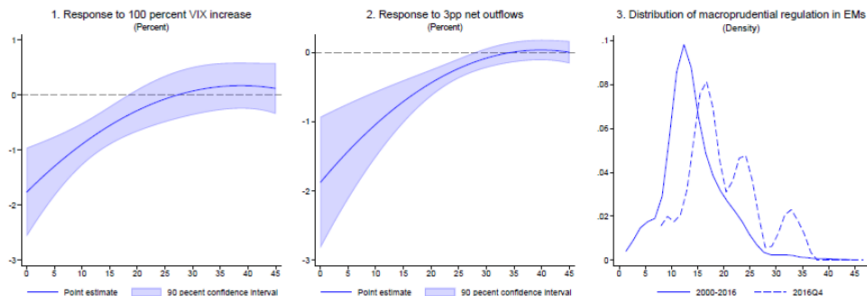
	All EMs	All EMs	All EMs	No fixed ER	All EMs
	(1)	(2)	(3)	(4)	(5)
Lag dependent variable	0.071 (0.057)	0.117 (0.088)	0.072 (0.050)	0.012 (0.054)	0.072 (0.048)
Lag output gap	-0.378*** (0.035)	-0.313*** (0.045)	-0.369*** (0.035)	-0.342*** (0.033)	-0.368*** (0.035)
Lag ln real GDP per capita	-0.880* (0.505)	-0.144 (0.440)	-1.689** (0.707)	-1.111* (0.630)	-1.713** (0.749)
Institutional quality	-0.281 (0.644)	-0.484 (0.455)	-0.132 (0.813)	-0.536 (0.938)	0.078 (0.930)
Linear trend	0.003 (0.005)	-0.002 (0.005)	0.005 (0.007)	0.002 (0.006)	0.002 (0.007)
Commodity terms of trade	0.050 (0.035)	0.056 (0.038)	0.037 (0.050)	0.083 (0.052)	0.035 (0.058)
US monetary policy shock	-0.158 (0.168)	-0.449** (0.189)	-0.008 (0.311)	-0.077 (0.318)	0.017 (0.357)
Ln VIX	-0.712*** (0.182)		-1.556*** (0.312)	-1.571*** (0.287)	-1.762*** (0.487)
Net outflows	-0.186*** (0.046)		-0.393*** (0.113)	-0.378*** (0.122)	-0.624*** (0.183)
Mpru			-1.380*** (0.401)	-1.480*** (0.404)	-1.514 (1.234)
US monetary policy shock * Mpru			-0.100 (0.109)	-0.074 (0.118)	-0.164 (0.221)
Ln VIX * Mpru			0.631*** (0.143)	0.609*** (0.148)	0.997** (0.419)
Net outflows * Mpru			0.108*** (0.036)	0.095** (0.043)	0.319*** (0.111)
MPru2					0.237 (0.209)
US monetary policy shock * MPru2					0.037 (0.037)
Ln VIX * MPru2					-0.129* (0.073)
Net outflows * MPru2					-0.040** (0.016)
Observations	2,260	2,260	2,260	1,658	2,260
Countries	38	38	38	32	38
F-statistic	73.1		33.1	24.4	18.6

Bergant, Katharina, Francesco Grigoli, Niels-Jakob Hansen, and Damiano Sandri (2020). "Dampening Global Financial Shocks: Can Macroprudential Regulation Help (More than Capital Controls)?" CEPR discussion paper 14948.

Decreasing effectiveness

- Dampening effect largest when macroprudential policy is low.

Figure 3: GDP Response in Emerging Markets to Global Financial Shocks



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SPIILLOVERS

- Macroprudential policy in a country affects bank lending from other countries.
- In addition, it can shift activity towards Non Bank Financial Intermediation (NBFI), i.e. shadow banking (Claessens et al. 2023).
 - Evidence on NBFI activity in 24 countries from 2002 to 2017.
- NBFI covers 5 blocks of activities with bank features. Rapid expansion in recent years.
 - Collective investment vehicles vulnerable to runs (investment funds). Largest component.
 - Lending with short-term funding (leasing firms).
 - Market activities with short-term funding (broker-dealer).
 - Facilitation of credit (credit insurance).
 - Securitisation-based credit (asset backed securities).

Effect of macroprudential on NBFi

- Index of net tightening of domestic macroprudential measures.
- Similar index for foreign countries: indices weighted by domestic country's holdings in the foreign countries.
- Effect of NBFi / total assets.
 - Domestic tightening raises NBFi ratio, foreign tightening lowers it.
 - Symmetric for tightening and easing.
- Assess effect of NBFi volume and bank volume separately.
 - Domestic tightening raises NBFi activity and reduces bank lending.

Effect on NBFIs volume and bank lending

- Tighter domestic measures raise **NBFI activity**, and reduce **banking activity**.
- Mirror effect of foreign measures.

Table 8. Effects of MaPs on the Level of NBFI Assets and Banking Assets

Explanatory Variables	Logarithm of NBFI Assets			Logarithm of Banking Financial Assets			Logarithm of Bank Deposits		
	(1) Other Countries' MaPs Weighted by Claims	(2) Other Countries' MaPs Weighted by Liabilities	(3) Other Countries' MaPs Weighted by Claims + Liabilities	(4) Other Countries' MaPs Weighted by Claims	(5) Other Countries' MaPs Weighted by Liabilities	(6) Other Countries' MaPs Weighted by Claims + Liabilities	(7) Other Countries' MaPs Weighted by Claims	(8) Other Countries' MaPs Weighted by Liabilities	(9) Other Countries' MaPs Weighted by Claims + Liabilities
LI.Net MaP Tightening	0.0602*** (0.0162)	0.0580*** (0.0163)	0.0562*** (0.0161)	-0.0257** (0.0112)	-0.0244** (0.0108)	-0.0231** (0.0110)	-0.0451** (0.0194)	-0.0435** (0.0186)	-0.0408** (0.0188)
LI.Net MaP Tightening, Other Countries	-0.0689** (0.0329)	-0.0469* (0.0247)	-0.0775*** (0.0297)	0.0215 (0.0186)	0.0765*** (0.0149)	0.0713*** (0.0167)	0.0303 (0.0334)	0.0901*** (0.0300)	0.0899** (0.0350)
Supervisory Authority Protection	-0.0863*** (0.0297)	-0.0823*** (0.0300)	-0.0822*** (0.0304)	-0.0533*** (0.0201)	-0.0603*** (0.0212)	-0.0574*** (0.0209)	-0.0932*** (0.0352)	-0.101*** (0.0332)	-0.0983*** (0.0341)
Other Controls ¹	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	260	260	260	260	260	260	259	259	259
Adjusted R-squared	0.989	0.989	0.989	0.996	0.996	0.996	0.989	0.989	0.989

Cerutti, Eugenio, Ricardo Correa, Elisabetta Fiorentino, and Esther Segalla (2017). "Changes in Prudential Policy Instruments - A New Cross-Country Database", *International Journal of Central Banking*, 13(1) pp. 477-503.

USE AND EFFECT OF CAPITAL CONTROLS

- Indices of capital account openness (such as Chinn-Ito) indicate whether controls are present or not (extensive margin), not whether they are tightening or loosening (intensive margin).
 - Annual data, based on de jure AREAER (annual report on exchange rate arrangements and exchange restrictions) announced by countries.
- Use instead a high frequency (monthly measure) of the Taxonomy of capital flow management measures, CFM (Binici, Das, and Pugacheva 2023).
 - Compiled by IMF staff since 2008.
 - Focus on measures that are “macro critical”, and include measures that may not be targeted at capital flows but have consequences for the balance of payments (e.g: currency specific reserve requirements).
 - Monthly measures, allow for the analysis of intra-year CFM usage.
 - 49 countries, including low income economies.

Measures of controls

- Count of changes in the measure: + 1 is a tightening, -1 an easing, on inflows (*IM*) or outflows (*OM*).
 - *NII*: net inflow index, sum of inflows tightening and outflows loosening.
 - *NOI*: net outflow index, sum of outflows tightening and inflows loosening.
- Measure of stance by taking the cumulative sum across months of *NII* – *NOI*.
- Measure intensity by weighting measures on specific categories of flows by the share of these categories in inflows or outflows.
 - Weighting by BoP capital flows, not by NIIP stocks (Aizenman and al. 2018).
 - With changing funding structure, NIIP measure can be highly lagging.
- Distinction between price based (reserve requirements, tax) and administrative (ban, quota) measures.
 - Price measures can work only if there is a well functioning financial market.

- Frequent use of measures, both on inflows and outflows. Loosening since 2015.

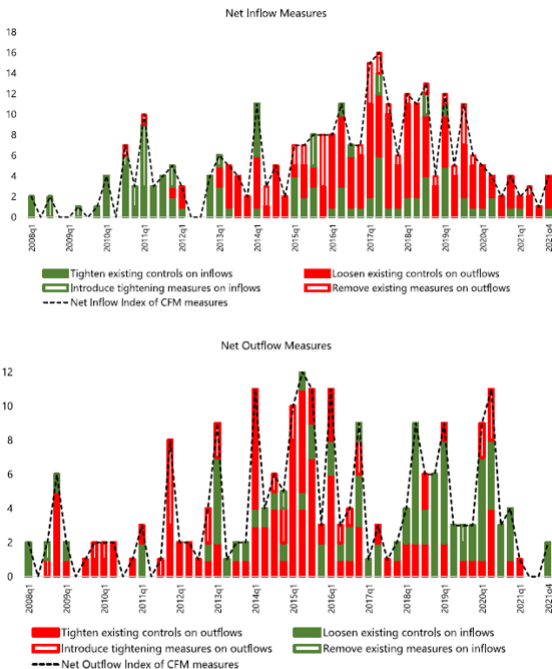
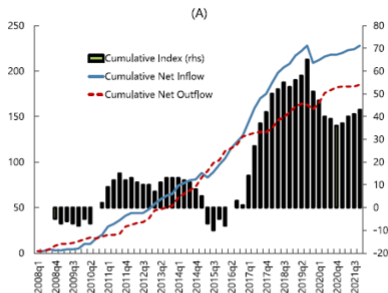


Fig. 2 Dynamics of the Net Indices of Inflow and Outflow Measures. *Source:* Authors' compilation from the Taxonomy

Binici, Mahir, Mitali Das, and Evgenia Pugacheva (2023). "The Incidence of Capital Flow Management Measures: Observations from a New Database" *IMF Economic Review* 72, pp. 441-486.

- Rising cumulative inflow tightening index since 2017. Evidence of reaction to house prices.



(B) Cumulative Net Inflow Index and House Prices

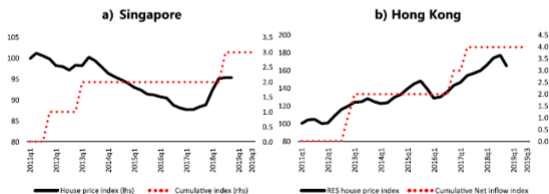


Fig. 4 Cumulative Net Inflow Index. The figure in Panel (A) is a simple aggregation over all countries in the sample. House price index 2011q1 = 100. *Sources:* Authors' compilation from the Taxonomy; IMF; OECD; Global Property Guide; Haver Analytics

Binici, Mahir, Mitali Das, and Evgenia Pugacheva (2023). "The Incidence of Capital Flow Management Measures: Observations from a New Database" *IMF Economic Review* 72, pp. 441-486.

Pattern of usage

- CFM are mostly used by emerging economies. They are subjected to volatile capital flows.
 - Use in advanced economies, mostly European periphery countries in the euro crisis (Cyprus, Greece, Iceland).
 - Limited use in low income countries. Capital flows can be mostly FDI, not “hot money”.
- CFM are used more by countries with managed exchange rates and not targeting inflation.
 - Consistent with trilemma.
- Price based measures (reserve requirements to limit carry trade) used on inflows, mostly in emerging economies.
- Administrative measures on outflows in low income economies. Also in some advanced economies (stamp duty on real estate purchases).
- Low income countries focus on outflows, other countries more even.
- CFM mostly on banking flows.
- Correlation with other measures very limited. Points to sensitivity of the sample and definitions.

- High frequency movements, but correlation with other indices very limited.

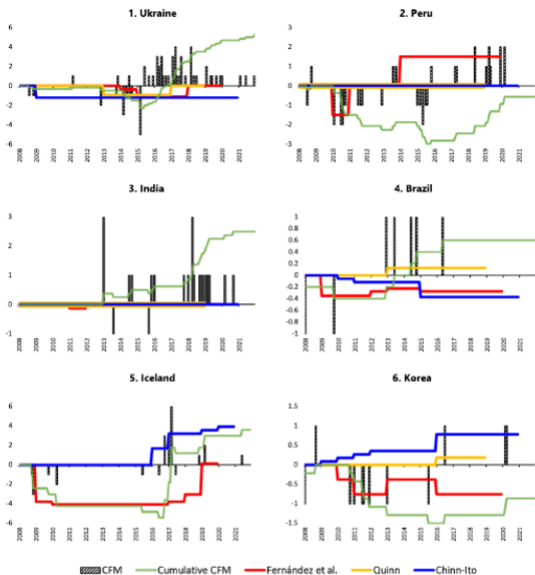


Fig. 7 Capital Account: Overall Convertibility vs. High-Frequency Recalibration. Notes: Positive values indicate more openness. CFM are at monthly frequency; comparators are at annual frequency. Red lines are the overall restrictions index from Fernández et al. (2016), inverted. Blue lines are the normalized capital account openness index from Chinn and Ito (2006). Yellow lines are the lack of financial restrictions on capital account transactions index from Quinn (1997). All lines (Cumulative CFM, Fernández et al., Quinn, Chinn-Ito) were rescaled to start at zero and fit within the same range. Thus, the values should not be interpreted as actual numbers, rather an indication of the direction of change in the series. Data from all sources reflects the most recent data updates. Sources: Authors' compilation from the Taxonomy; Fernández et al. (2016); Chinn and Ito (2006); Quinn (1997)

Binici, Mahir, Mitali Das, and Evgenia Pugacheva (2023). "The Incidence of Capital Flow Management Measures: Observations from a New Database" *IMF Economic Review* 72, pp. 441-486.

IMPACT ON FLOWS

- Panel analysis of impact of CFM on capital flows on annual data (Chen 2024).
- De jure measure (AREAER). Capital outflow controls sum restrictions on outflows and allowance on inflows.
 - Three regimes of capital mobility: close, middle, open.
- Capital flows: portfolio and banks.
- Distinguish duration of controls. Long if controls in place for more than 90% of the time (1995-2015), short otherwise.

Impact on outflows

- Controls **reduce outflows**, even with other variables. **Full controls (close)** are needed for an effect.
- Only **short controls** have an effect robust to other variables.

Table 2
Aggregate capital control index on capital outflows.

	(1) Outflow/GDP	(2) Outflow/GDP	(3) Outflow/GDP	(4) Outflow/GDP	(5) Outflow/GDP
Capital Control Index	-10.88*** (3.217)			-5.134* (2.603)	
Close		-9.166*** (3.003)			
Mid		-1.782 (4.621)			
LongClose			-9.247*** (2.690)		-2.665 (2.455)
ShortClose			-6.934*** (2.168)		-5.385*** (2.003)
Real GDP per capita				3.526 (3.523)	3.648 (3.678)
Growth of GDP per capita				-0.259 (0.247)	-0.274 (0.253)
Institutional quality index				-0.0512 (0.129)	-0.0453 (0.131)
Private credit to GDP				0.138*** (0.0433)	0.138*** (0.0447)
Domestic real interest				-0.0937 (0.0609)	-0.0829 (0.0595)
Constant	14.12*** (3.103)	13.19*** (3.368)	12.39*** (2.583)	-24.37 (25.73)	-26.81 (26.92)
Year F.E.	YES	YES	YES	YES	YES
N	1802	1802	1802	1231	1231
R ²	0.054	0.051	0.051	0.139	0.138

Note: *LongClose* is a country dummy variable for countries with long-standing comprehensive controls on capital account. *ShortClose* is the dummy variable for episodic comprehensive controls. *Mid* represents cases of limited controls. Standard errors in parenthesis and clustered by country. * $p < 0.1$, * * $p < 0.05$, * * * $p < 0.01$.

Chen, Gao (2024). "Capital controls and capital flows: Do controls reduce the size of flows?" *Journal of International Money and finance* 145.

- Results robust to using propensity score matching: similar countries, but only one with controls.
- Effects are smaller since 2008, but still significant.
- Results are robust to looking at all outflows, including FDI.
 - Effect also on overall inflows, but only in emerging economies.
- Effect on controls on equity or debt not robust to other variables, but effect of controls on banks is.

Taking stock of the literature

- Surveys of recent theoretical and empirical work on controls (Ma and Rebucci 2019; Erten, Korinek and Ocampo 2021).
 - Capital controls built on the resident - nonresident distinction, different from currency-based measures.
- Capital controls can be of use for three theoretical reasons:
 - Pecuniary externality: presence of asset prices in occasionally binding constraints.
 - Demand externality: adverse shock and monetary policy stuck at the zero lower bound. Includes restoring policy autonomy in a trilemma.
 - Terms-of-trade manipulation (akin to optimal tariff)
- If capital controls can work, other tools can also (macroprudential).
 - Usefulness of capital controls can depend on the details of the model. Concern about robustness.
 - Capital controls may not be enough on their own (e.g.: if the borrowing constraint depends on an asset price other than the exchange rate).
 - Other tools can be better to deliver the required sectorial re-allocation of demand.

Empirical evidence

- Two broad strands of work: construction of databases with measures of capital control usage (also macroprudential); assessment of impact.
- Pre-crisis view was that controls can affect the composition of flows, but not the overall volume or the exchange rate.
- Evidence that capital controls can affect the structure of credit (less credit from abroad), but not its volume.
 - Possible side effects: lower foreign currency borrowing from banks, but higher directly from firms.
 - Controls can help with policy autonomy, and can reduce the likelihood of crises.
- Views on capital controls have evolved.
 - IMF viewed them as not appropriate before the Asian 1997-98 crisis.
 - View has changed: now seen as part of the toolkit, as a last resort measure.

INTEGRATED POLICY FRAMEWORK

- Large expansion of the policy toolkit in recent years: macroprudential policies, FX interventions, capital controls.
- Analytical models tend to focus on a subset of these policies, as well as specific shocks.
- What is the proper conduct of policy? What tools are adequate, why, and when?
- Integrated Policy Framework: research program to develop models encompassing many tools and many shocks.
 - Identify the relevant misalignments.
 - Identify which tools can handle misalignments, and under which circumstances.
 - Understand better how tools interact. No “one tool for only one target” view.
 - Challenge: get some broad lessons out of a very long list of specific situations.

A model of ex-ante and ex-post policies

- Framework that considers crisis and normal periods, contrasting policies before shocks happens to policies once they do (Cavallino and Hofmann 2022).
- Features of relevance to emerging economies.
 - Dominant currency pricing: full transmission of exchange rate movements to import prices, no transmission to foreign price of the country's exports.
 - Investment through banks, occasionally binding leverage constraints.
 - Shallow foreign exchange market: endogenous deviation from UIP. Opens a role for foreign exchange intervention.
- Three periods.
 - Period 0: positive shocks (such as capital inflows), can lead to the buildup of vulnerabilities. Ex-ante policy actions, with an high of exposure to tail risk in period 1.
 - Period 1: adverse shock (sudden stop), can lead to a binding leverage constraint with severe macro consequences, especially when leverage set in period 0 is high. Ex-post policy actions.
 - Period 2: long run, with prices adjusting.

Main features

- Real elements.
 - Household consumes domestic and imported good.
 - Firms produce traded good, sold domestically and abroad, with capital and labor.
 - Capital produced by specific firms that need to borrow from banks.
- Financial elements.
 - Households invests in bank deposits.
 - Banks lend to capital firms and buy government bonds. Can be subject to a leverage constraint.
 - Foreign investors can buy government bonds. Higher capital flows absorb government bonds, so banks lend more to firms and leverage less.
 - Central bank buys government bond and foreign bond, financed by money.

- Uncovered interest parity condition (E is the exchange rate, R^b the domestic interest rate, R^* the foreign interest rate, and Π inflation (rates are measured in unit of consumption basket):

$$1 = \mathbb{E}_t \left[\frac{E_t}{E_{t+1}} \frac{R_{t+1}^b}{\Pi_{t+1} R_t^*} \frac{\exp(-\tau_t^b)}{\eta_t^*} \right]$$

$$\eta_t^* = \left[\exp(\varepsilon_t^\kappa) + \varpi \left(\exp\left(\frac{P_t^b B_t^* - P^b B^*}{P_H Y_H}\right) - 1 \right) \right] \exp((\varsigma + \zeta_b) \mu_t)$$

- The first two ratios are the **usual UIP**. τ_t^b is a tax on inflows.
- η_t^* is an **external premium** with stochastic component $\exp(\varepsilon_t^\kappa)$, sudden stop shock, and **shallow market effect** (if $\varpi > 0$). If the amount of bonds that foreigners need to hold, $P_t^b B_t^*$, lower, the premium is reduced.

- In period 1 the constraint can be binding or not. The expected value of variables is thus not zero, due to non-linearity across these regimes.
- Policy in period 0 takes account of the risk of facing a binding constraint.
- Menu of policies:
 - monetary policy: short term interest rate.
 - macroprudential policy: tax / subsidy to loans by banks to firms.
 - CFM: tax / subsidy on foreigners' holdings of government bonds.
 - Bond market policy (QE): sell / buy government bond in exchange of money.
 - FX intervention: sell / buy government bond in exchange of foreign currency bonds.

- Exogenous tightening of external premium ($\varepsilon_t^\kappa > 0$).
- Without binding constraint, banks buy the bonds sold by investors, and leverage more.
 - No impact on output, inflation because of exchange rate depreciation.
- Different patterns if bond buying by banks hits the leverage constraint.
 - Constraint purchases of bonds, leads to higher interest rate.
 - Lower asset prices, erodes banks' equity, makes the leverage constraint even tighter.
- Compute the optimal policies from the model.

Ex-post policy response

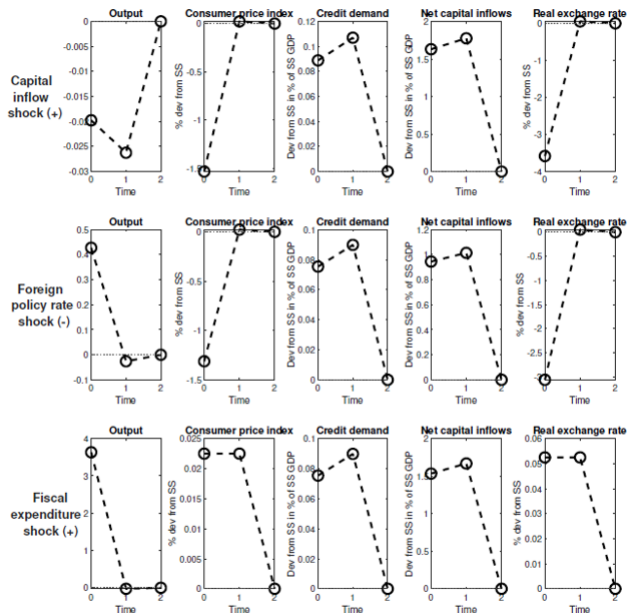
- Monetary policy.
 - If the constraint is not binding, increase the interest rate.
 - If it binds, trade-off between recession and inflation. Lower interest rates raises capital price and helps net worth of banks, but it reduces further capital inflows.
- Macroprudential policy and CFM.
 - Subsidize lending to limit the credit crunch.
 - Lower tax on capital inflows to dampen the shock, and limit the exchange rate depreciation.
- QE and FX intervention.
 - Buy government bonds from foreigners, limit the adjustment that banks need to make.
 - With shallow FX market, intervention limits the depreciation and absorbs government bonds, thus relaxing the leverage constraint. (as does QE, which is more direct).

Ex-ante policy

- In period 0, policy takes account of the tail risk of a sudden stop in period 1. Keep banks away from the binding leverage constraint.
- Monetary policy leans against the wind. Higher interest rate reduces lending and leverage, and attract foreign investment in gvt. bonds.
 - Moderate credit restriction to reduce risk of future crisis.
 - Public debt plays a role: a higher debt in period 0 leads to higher leverage, hence more tail risk in period 1.
- Macroprudential tax reduces loan demand (and leverage). It also increases savings (and leverage), but this is secondary.
- CFM reduce foreign funding of government bonds, leading to more bank leverage. They also reduce loan demand and leverage. Can mitigate tail risk, but less than macroprudential tax.
- QE has no effect: bonds and money are substitutes. FX intervention (with shallow market) to accumulate reserves and reduce tail risk.
- Shocks that increase tail risk (higher borrowing demand) leads to tighter policies.
- Pressure on monetary policy is lowered by using the other tools.

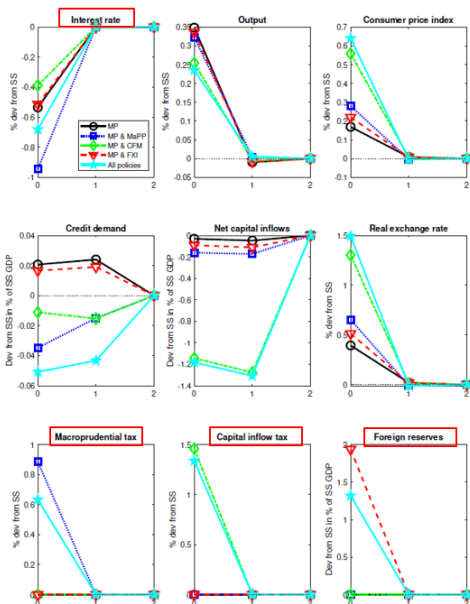
- Beneficial shock at time 0 (inflows, lower foreign interest rate), lower inflation, appreciate the exchange rate, boost credit. Lower rate raises output (moderate decrease from inflows).
- The raise tail risk.

Figure 3.1: Time-0 Shocks: Impulse Response Functions



- Following inflows: lower interest rate (but raises credit), tighter macroprudential and CFM, higher reserves.
- Credit affected by macroprudential and CFM, exchange rate by CFM.

Figure 3.2: Time-0 Policies: Capital Flow Shock (+)



Cavallino, Paolo, and Boris Hofmann (2022). "Capital flows and monetary policy trade-offs in emerging market economies", BIS working paper 1032.

- Following lower foreign rates: tighter macroprudential and CFM, higher reserves.
- Higher interest rate, unless macroprudential is also used.

Figure 3.3: Time-0 Policies: Foreign Monetary Policy Shock (-)

